Exercise 1:
New coal mine in Kalimantan (static simulation)

*Although this is a dynamic course, we will start with a simulation using the static version of IndoTERM. The aim of this exercise is to introduce to you the many files (simulation ingredients) used in simulations: model equations (TAB); model data (HAR); command files (CMF) and simulation results (SL4). **Do not worry if you do not understand everything straight away**. In Exercise 3 we will review all the simulation ingredients again and introduce RunDynam, the software we use to run dynamic simulations.*

*A large international mining company, **Ecologia**, has discovered new coal seams in East Kalimantan and plans to build a new coal extraction facility. The exploration of new coal deposits in Kalimantan is modelled in two phases; the "construction phase" which is simulated using a short-run closure (Part A) and the "operational phase" which is simulated using a long-run closure (Part B). New export markets are already secured, which will absorb the output of the new mine without affecting export prices received by existing mines. Assess the regional and national effects of the project, using a static version of IndoTERM.*

Part A: The construction phase in developing a new coal mine in Kalimantan

1. Getting Started

We assume that you have already installed the GEMPACK and RunDynam software on your computer, and that you have already copied other course material into the folder C:\DTERMCRS. That folder should contain the following subfolders:

- *EX1 EX2 EX3 and EX45:* These folders will be used for exercises 1 to 5.
- *ZIPS*: This folder contains stored RunDynam simulations which you will use in Exercises 3-5.
- *PDF*: This folder contains all the documents in your course folder, and some other documents.
- You may see some other subfolders -- these are optional or not needed yet.

During the dynamic TERM course you will work in one of the folders, **C:\DTERMCRS\EX??**

Good luck and let's get started!

You will be working (editing files, running simulations, looking at results) in the **working directory** of **C:\DTERMCRS\EX1**

In Part A we are going to simulate the short run impact of an increase in investments in the coal industry located in East Kalimantan. We have included a short article about the mining industry in Kalimantan – see next page.

Work through this exercise and **write down** answers to the questions onto the instruction sheet. The questions are scattered through the instructions.

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MINING AND RESOURCES

Kalimantan Coal: The Second Phase

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FILED UNDER ASIA, COAL, KALIMANTAN, MINING, RESOURCES

Indonesia has progressed in the past 12 years from exporting a paltry 20m tonnes of coal in 2000, to become the world largest exporter of Coal. Australia was until 2012 1st and China 2nd.

Indonesian coal has become more attractive to international buyers as the price is much lower than Australian coal, and the costs for shipping and time required to ship are much lower.

Known coal deposits of over 35 billion tonnes of thermal coal and coking exist and exploration continues to find more.

Coming off a low export base Indonesian exports have risen over the past 10 years from 40 mtpa to 114 mtpa in 2011 and this continues to rise. 2012 was over 200 mtpa. The number of active mines in Indonesia has grown massively in the past 10 years from 40 mines to over 100, with many Australian companies seeking to mine and export coal.

India has traditionally been the largest buyer of Indonesian coal, but were surpassed recently by China. Korea and Vietnam are also large buyers of Indonesian Coal.

Indonesian rules for export state that a percentage of all coal mined must be sold to the domestic market. Domestic Market Obligations.

Bulk export infrastructure is lacking with little rail infrastructure and due to the geography of the region many shallow rivers with few deep water ports. The rivers have proved both a help and a hazard. The rivers have assisted many companies to load the coal onto barges, which can be taken to deep water and then transferred to other vessels or in some cases the barge is taken directly to other customers in Malaysia and Vietnam.

This also had hindered large scale high volume export as the coal must be transshipped and handled multiple times. Coal is often loaded on trucks then taken to a river, shipped by barge to a waiting ship out at sea and then craned onto the ship. To fill a panamax 200,000 tonne ship could take 2-3 days whereas a deep water port with specialized loading facilities could load the ship in under 24 hours.

There have great improvements in transshipping though which improve loading rates. Additionally there is strong interest in building deepwater port facilities and rail lines to transfer the coal.

MEC Coal are looking to build a heavy haul rail line to transport coal and there is another Russian based conglomerate looking to build rail.

There are many different deepwater bulk port facilities proposed.

Indonesia have recently passed new laws which limit foreign ownership of new mines to 49%. These laws announced 9th March 2012, but passed over a month before, have left current mine ownership in question with conflicting reports by different ministers on the impacts for current ownership structures. Major changes to mining legislation, ownership and operation of both existing and new mines. Whilst the laws have been passed there has been little further action and many international mining companies are unaware of what impact these will have or if they will be imposed.

Kalimantan Coal 2013 will analyse developments within the Indonesian coal legislation, regional issues, business opportunities and advances in coal markets while simultaneously addressing issues such as infrastructure, land use agreements and environmental impact assessments. Find out more about the conference here.

2. Opening a DOS box and finding the right folder

The command prompt, or "DOS box" is a small, but important, part of using GEMPACK. In this section we explain how to open a DOS box and navigate to the course work folder.

The DOS box (also called the *command prompt*) is the traditional way of interacting with computers, used before the mouse was invented, and still used today by computing professionals. It remains the quickest way to perform some operations.

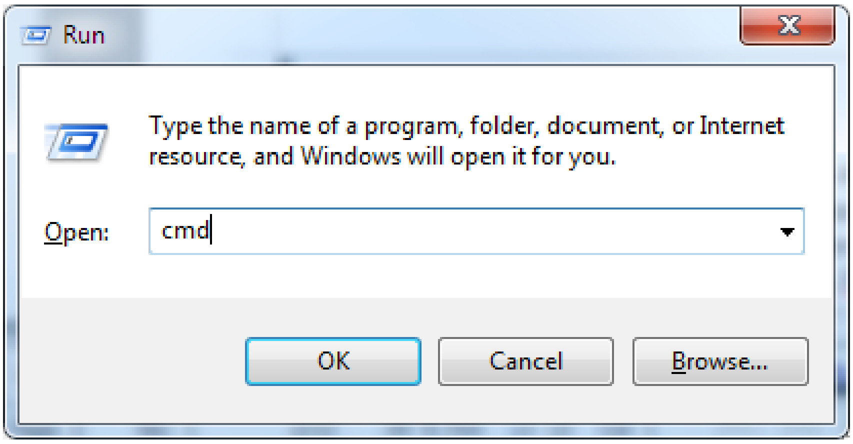
2.1 Opening a DOS box

If you already know how to open a DOS box, do it now, and jump to the next subsection, *Navigating to the work folder*.

There are many ways to open a DOS box -- but some only work with some versions of Windows. Here is one method that works with with most versions.

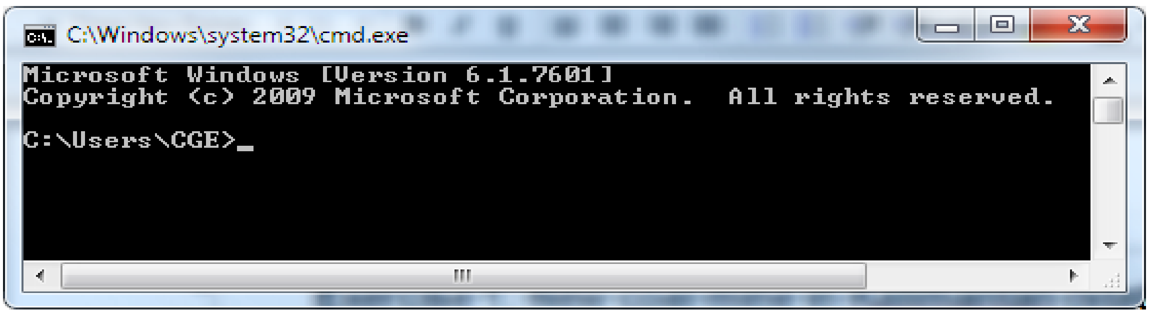
Locate the Windows key near bottom left of your keyboard. It may look like this  or this .

Simultaneously press the Windows key and the "R" key. The Run box should be displayed:



Type in "cmd", then click OK.

You should see a DOS box rather like this:

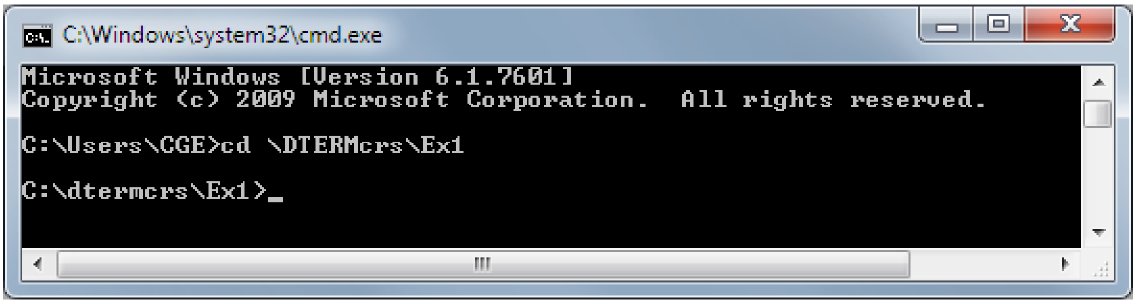


2.2 Navigating to the work folder

Type the following line, exactly as shown below, into your DOS box, then press Enter:

```
cd \DTERMCRS\EX1
```

The DOS box should now resemble:



Now type:

```
dir
```

to list the files in this folder.

2.3 Other ways to open a DOS box in a particular folder

There are many other, quicker, ways to open a DOS box in a particular folder.

For example, you could launch Windows Explorer by simultaneously pressing the Windows key and the "R" key. , Then, if you right-click on a folder, *while pressing the Shift key*, you can select the option **Command Prompt Here** -- which opens a DOS box in the selected folder.

Most of the GEMPACK Windows programs offer a menu command to open a DOS box.

- In TABmate: **File.|.DOS box in current folder**
- In RunDynam: **File.|.Shell to DOS in working directory**
- In ViewHAR and ViewSOL: **Programs.|.DOS prompt**

3. Reviewing the TAB and data file

3.1 Viewing the TAB file

In the DOS box type:

```
tabmate term.tab
```

to see the model TAB file. Browse through the tab file. What is its purpose of TERM.TAB?

The TAB file contains the model theory of the IndoTERM. It is written in TABLO language. The code is divided into Excerpts – each excerpt explains a specific theme or contains specific information. For example, Excerpt 1 defines all the files and sets used in the simulation.

Various colours appear on the screen for different parts of the TAB file; for example, TABLO Keywords are in BLACK, comments are in royal blue and so on[1]. The first page consists of a several comments (in blue) describing recent changes to the model.

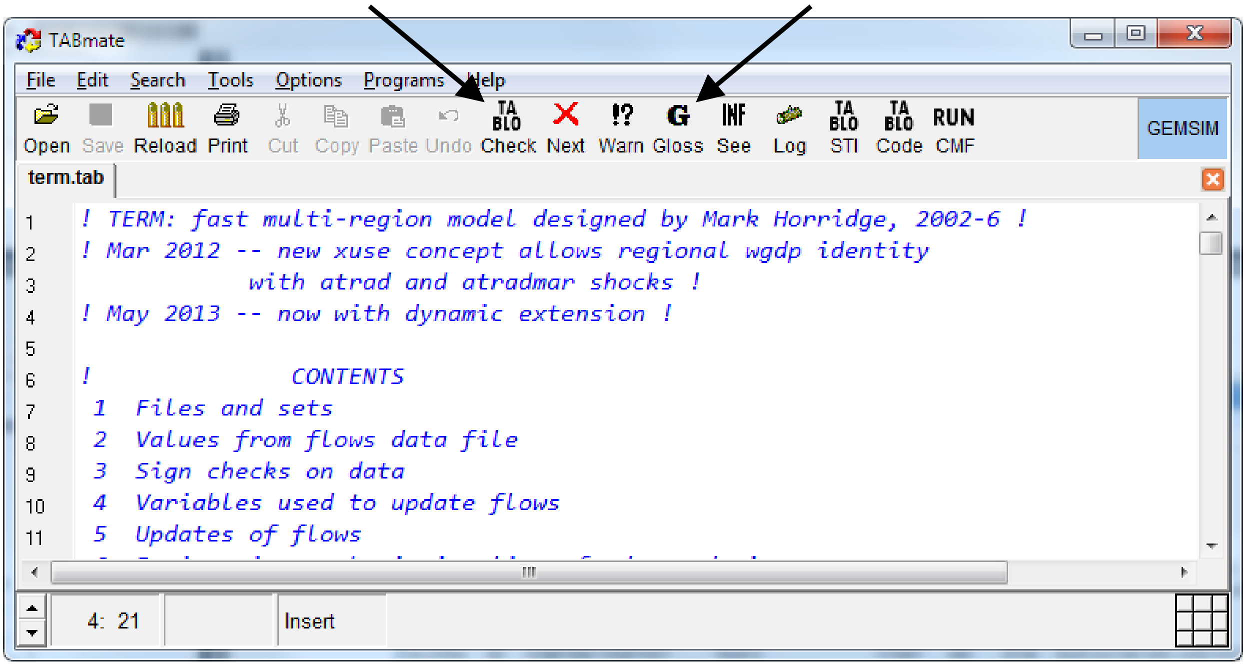
Scroll down a page and you will see statements describing the three files used by this model:

- the input file INFILE containing input-out data and elasticities,
- the file REGSETS containing the sets and mappings used in the model, and
- the output (new) file SUMMARY which contains summary and checking data calculated from INFILE[2]

[1] If this is not the case you can restore the default properties, (Select menu item Edit | Restore Default Properties) or you can change the screen properties by selecting Edit | Properties to set the font, the font size and various colours on the screen.

[2] INFILE and SUMMARY are 'logical' file names. When you run a simulation, you have to specify the name and location of corresponding actual files.

Place your cursor on the word INFILE and press the **Gloss** button (at top middle of the screen). If this gives a message saying "*no info available: run Check to generate*" at the bottom of the screen[3], click on the button marked **TABLO Check** (to left of *Gloss*) and then try **Gloss** again.



This should show you all places in the TABLO Input file where the file INFILE is mentioned -- all the places where initial data is read from file[4]. Click on the line number at the start of the Gloss line and TABmate will take you to that line[5]. A related feature is: if you hold the Alt key down, while moving the mouse over green words in the TAB file, you should see a short definition of each symbol.Return to the top of the file (Ctrl and Home goes to the top of the file, Ctrl and End to the bottom of the file) and

Search | Find | Search forward from top

for the variable name **atot**.

```
(all,i,IND)(all,d,DST) atot (i,d) # All-input-augmenting technical change #;
```

The "(all,i,IND)" means that atot is a matrix variable with one value for each INDustry in each region of use (DEStination region). Click on **atot** and press the ***Gloss*** button (at top middle of the screen). You can see that atot appears in 5 equations: E_xint_ s, E_xprim, E_pvar, E_pcst and later on the long E_contincind equation. Click on the red line number at left of equation E_xprim and you should see:

```
E_xprim # Use of composite primary factor # (all,i,IND)(all,d,DST)
xprim(i,d) = xtot(i,d)+ atot(i,d) + aprim(i,d);
```

Each of the variables xprim, xtot, atot and aprim is a percentage change. The "a" variables atot and aprim are technological change variables, normally exogenous (values fixed outside the model). Suppose output were fixed (xtot=0), a shock of 10% to atot("Paddy", "Kalimantan") would mean that for the Paddy industry in Kalimantan, the values of xprim("Paddy","Kalimantan") must also increase 10% to keep the equation balanced.

Press ESC or spacebar to close the Gloss window.

---

[3] If the message disappears too fast for you to read, click on the line at the bottom of the screen and the message will reappear.

[4] You can click on any variable, coefficient or set and press the Gloss button to display a list of every statement in the file mentioning that symbol. For a variable, say, the first of these statements will usually furnish a definition. The remainder show how the variable is used. Line numbers accompany each statement; you can click on these to jump to that location in the TAB file. If you press Gloss when the cursor is not on a variable, coefficient or set, you get a different list showing the definition of each variable, coefficient or set mentioned in the current statement.

[5] You can return to your original place as follows: At bottom left, the Location Indicator panel shows current line and column numbers. Click there and a window appears, listing lines that you jumped to or from. The most recently visited line appears at the top. You can click on any of these lines to jump there.

Exit from TABmate in the usual Windows way by **File | Exit.** (There are usually alternatives in terms of keystrokes instead of the mouse action. For example you can use keystrokes ***Alt*** followed by ***F*** followed by ***X*** in order to exit.)

3.2 Viewing the data directly using ViewHAR

The input-output data used during this course for the TERM model is contained in the data file AGGMOD.HAR. To view AGGMOD.HAR using a special viewing program called ViewHAR, type in the DOS box the following command[6]:

```
viewhar aggmod.har
```

Each of the rows corresponds to a different array of data on the file. Look at the column under the heading **Name** to see what these arrays are:

aggmod.har in C:\DTERMcourse\Exercises\E1_MineStatic

File Contents Edit Sets Export Import History Search Aggregation Programs Help

	Header	Type	Dimension	Coeff	Total	Negs	Min	Name
1	COM	1C	41 length 12					Set COM Commodities
2	MAR	1C	6 length 10					Set MAR Margin coms
3	IND	1C	41 length 12					Set IND Industries
4	OCC	1C	4 length 12					Set OCC Skills
5	REGD	1C	12 length 11					Set DST Regions of use
6	REGS	1C	12 length 11					Set ORG Regions of origin
7	REGP	1C	12 length 11					Set PRD Regions of production
8	MAKE	RE*	COM*IND*DST	MAKE	5813164	0	0	MAKE matrix
9	BSMR	RE	COM*SRC*USR*DST	USE	6621781	0	0	Basic + margin flows
10	UTAX	RE	COM*SRC*USR*DST	TAXES	115967	480	-9138	TAXES(COM:SRC:USR:DST)
11	TRAD	RE	COM*SRC*ORG*DST	TRADE	6003732	0	0	sourcing matrix
12	MARS	RE	MAR*ORG*DST*PRD	SUPPMAR	618050	0	0.000	margins supplied by PRD on goods passing fr
13	TMAR	RE	COM*SRC*MAR*ORG*DST	TRADMAR	618049	0	0	margins on trade matrix
14	2PUR	RE	COM*IND*DST	INVEST	693007	0	0	investment at purchasers prices
15	STOK	RE	IND*DST	STOCKS	23200	153	-1336	Domestic inventories

Double-Click on an item to view it (or arrow keys + space bar)

The first item, COM, is the list of commodities in this database. The array is of type 1C which means an array of strings. Double click on COM to see the commodity names. To return to the Contents Screen, click twice on any cell [or click on **Contents** in the ViewHAR menu].

How many commodities and industries are there ? *41 of each*

How many regions? *12*

Find the header 1CAP. Write down the value for the East Kalimantan coal industry. You will use this value later in this exercise. *19,477*

Study the regional map (a few pages forward) and identify the region pertaining to the simulation.

In aggmod.har, find the matrix called USE (search for the header "BSMR"). This matrix shows the use of commodity c from source s by user u in region d. Since it is a 4-dimensional matrix, you cannot see *all* of it at once. Instead you may choose to see:

- a subtotal -- obtained by adding up over all commodities, sources, users or regions.
- a "slice" -- which shows values for just one commodity, source, user or region.

This is controlled by the drop-down list boxes at top right. By viewing the USE matrix, what is the value of domestic coal, exported from Papua? You may adjust the drop-down boxes at the top right to show:

The value of domestic coal, exported from East Kalimantan (EKalimantan)? *35,339*

[6] You can use uppercase or lowercase letters, as you like.

ViewHAR can present numbers as shares. To see *Row shares*, look at the box in the top left hand corner beside the green and yellow bands. Click on this box and choose ***Row*** . You should see shares that add to 1 down each row. Switch off the Shares View, by again clicking on the Shares box -- this time choose **None**.

What is the share of domestic coal, exported from East Kalimantan? *88%*

Close ViewHAR by choosing **File | Exit.**

4. Viewing the CMF file

The file **TERMSR.CMF** contains the simulations details, including shocks to investment in the coal industry in Eastern Kalimantan. In the DOS box, type

```
tabmate termsr.cmf
```

to see the closure and shocks for this simulation. The main bits of information in this CMF file are:

- the model to use, in the line:

  ```
  auxiliary files = TERM;
  ```

- the actual file names, AGGMOD.HAR, AGGSETS.HAR and TERMSUM.HAR, that correspond to the logical file names, INFILE, REGSETS and SUMMARY, which are mentioned in the TAB file. By default, the solution file is named after the CMF file -- so in this case the solution will be stored in file termsr.SL4;

- the solution method: Euler 6;

- the closure, or list of exogenous variables. The model can determine the value of most *but not all* variables. Some variables must be held fixed (or shocked) by the modeller. These are called *exogenous*. The choice of *which* variables are to be exogenous varies between simulations. In this simulation, industry capital stocks (variable xcap) are held exogenous. The fixed capital stocks identify the simulation as short-run;

- swap statements to create a short-run closure;

- swap statements related to the investment project;

- the shock is specified at the end of the file. In this simulation we shock the variable xinvitot(“coal”,”EKalimantan”) to increase by 100%. These shocks are used to simulate the main effect of an increase in investment for the coal industry in East Kalimantan.

Identify the "exogenous" variables and "swap" statements which specify that the closure is similar to a short-run closure (capital stocks fixed, investment fixed nationally, real wage fixed) . Explain what the following two swap statements do. [Hint: search for NatMacro(*"RealInv"*) and fhou in TERM.TAB]

```
swap invslack =  NatMacro("RealInv");
swap xhouhtot =  fhou;
```

The first swap statement makes aggregate investment exogenous and allows a slack variable to adjust. The second swap statement makes regional consumption follow wage income.

In the policy-specific swap statements, what does the following statement do? Why do we want xinvitot(*"coal"*,*"EKalimantan"*) exogenous?

```
swap finv1("coal","EKalimantan") =  xinvitot("coal","EKalimantan");
```

This swap statement renders investment in the coal industry in Eastern Kalimantan exogenous. This allows us to introduce shocks to xinvitot(“coal”,”EKalimantan”) directly, simulating the impact of an increase in investments for this industry.

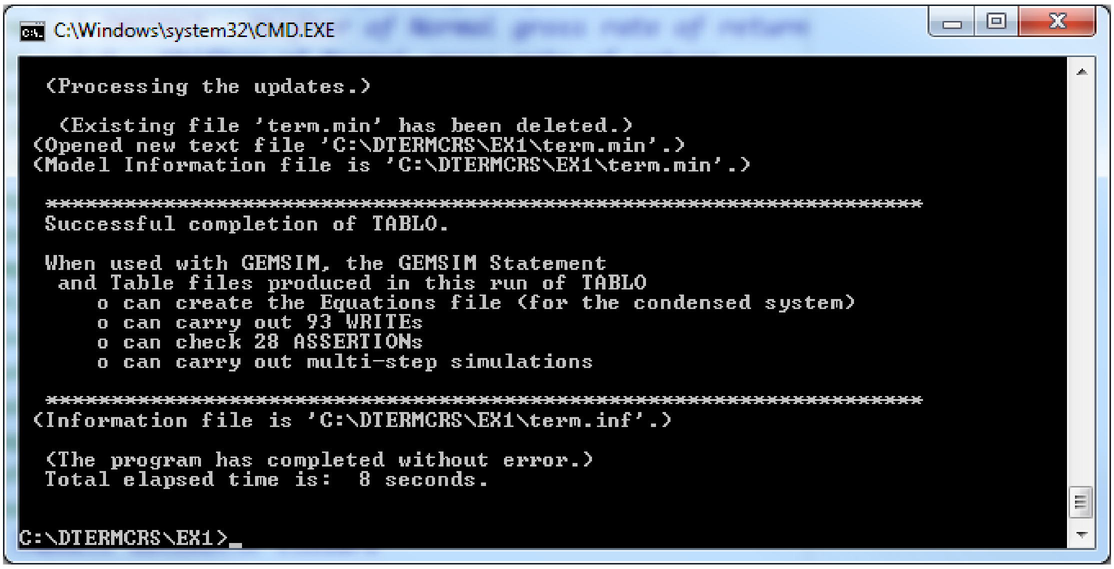
5. Running the simulation from a DOS box

The DOS box should still be open. If not, go to TABmate, and click on **File | DOS Box in current folder** to open a DOS box in the **C:\DTERMCRS\EX1** folder.

Carefully type in the following:

```
tablo -pgs term
```

and press Enter. You should see many messages flashing past, ending (after about 10 seconds) with something like:



If instead you seem to see an error message, please consult an instructor.

To run the simulation, you need to specify the corrent command file. In the DOS box type:

```
gemsim -cmf termsr.cmf
```

The program should complete without error. If the program shows an error message, please consult an instructor. To view the results, type the following command in theDOS box:

```
viewsol termsr.sl4
```

to see the results. A Contents page, listing variables should appear.

6. View simulation results using ViewSOL

The ViewSOL Contents screen shows the names of the variables, sorted alphabetically. To see the values of a variable, double-click on its name. To return to the Contents screen, double-click on any number (or select ***Contents*** in the ViewSOL menu).

Start by locating and examining values of in the **Macro** tab. This is just a convenient summary of national scalar results. All of the Macro results are measured in percentage changes[7]. Values for exogenous variables are shown in red (eg, invslack). Scroll down the list to find price indices (first letter "p"), nominal values (first letter "w"), and quantity indices (first letter "x").

No return to the **NatMacro** variable and complete the table below:

[7] Later you will see ordinary change variables (their names often start with “del”).

		Column A
	Main macro variable	**Short-run**
1.	RealHou: aggregate household spending	*0.16*
2.	RealInv: aggregate investment	*1.22*
3.	RealGov: aggregate government spending	*0*
4.	ExpVol: aggregate exports	*-0.58*
5.	ImpVolUsed: aggregate imports	*0.42*
6.	RealGDP: real expenditure side GDP	*0.09*
7.	AggEmploy: aggregate employment	*0.16*
8.	AggCapStock: aggregate capital stock	*0.0*
9.	ExportPI: export price index	*0.11*
10.	CPI: consumer price index	*0.21*
11.	p1lab_io: average nominal wage	*0.21*
12.	realwage_io: average real wage	*0.00*

What happened to national real investment [Look at the closure].

In the short run closure aggregate investments is held constant. We endogenise aggregate investment to allow it to increase due to the investment in East Kalimantan.

Double-click on any number to return to the Contents screen. Then find the **MainMacro** variable and examine its values. MainMacro is the regional version of NatMacro. Complete the table below.

	Variable	**East Kalimantan**	**North West Java**
1.	RealHou: aggregate household spending	*2.10*	*0.01*
2.	RealInv: aggregate investment	*10.92*	*0.01*
3.	ExpVol: aggregate exports	*-1.26*	*-0.48*
4.	ImpVolUsed: aggregate imports	*3.30*	*0.14*
5.	RealGDP: real expenditure side GDP	*0.90*	*0.01*
6.	AggEmploy: aggregate employment	*2.10*	*0.01*
7.	AggCapStock: aggregate capital stock	*0.0*	*0.0*

Double-click on any number to return to the Contents screen. Then scroll down till you find the variable **xtot** (industry outputs by region). Double-click to view the numbers.

Which three non-land using industries in East Kalimantan are most negatively affected? [Note: the land-using industries are the first 10 industries – from Paddy to OtherMining].

Shoes, EdibleOil, FishProds. A large share of the output of these industries is exported.

Which three industries expand the most and why?
Construction (1.02%), Cement (0.83%) and FabMetalPrd (0.29%). Their output is mainly used in the construction industry which is mainly used for Investment.

Return to Contents. Note down below the following two results -- you will need them later.

- xcap("Coal","EKalimantan") = *0 (exogenous)*

- pcap("Coal","EKalimantan") = *-0.61*

7. Other output files

7.1 The Summary file

In the DOS box type

```
viewhar termsum.har
```

And open the *SUMMARY* output file called TERMSUM.HAR. It contains various useful tables and data summaries which are calculated only from the initial input data file AGGMOD.HAR[8]. Experienced modellers rely on a good knowledge of the main features of their database. To understand why some industries perform better than others in a simulation, we have to know the special characteristics of each sector. The SUMMARY file contains various data that have proved useful in the past. Some of the most useful are:

Header	Dimension	Coeff	Name
PCST	IND*FAC*DST	PRIMCOST	Total factor inputs
CSTM	IND*COSTCAT*DST	COSTMAT	Cost Matrix
VTOT	IND*DST	VTOT	Total industry cost plus tax
LUSE	COM*SRC*DST	LOCUSE	Non-export delivered value of regional composite c,s in d
MSHR	COM	IMPSHR	Import share in non-export use
GNSM	DST*GDPINCCAT	GDPINCSUM	Income GDP breakdown
GESM	DST*GDPEXPCAT	GDPEXPSUM	Expend GDP breakdown
NGSM	NATGDPEXPCAT	NATGDPEXPSUM	Expend GDP breakdown
SHXP	COM*REG	EXPSHR	Share of output eventually exported
VMNU	COM*SRC*MAINUSR*DST	VMAINUSE	Simplified USE matrix
VMDS	COM*REG*MAINDEST	VMAINDEST	Main uses of dom com c made in r
NMDS	COM*MAINDEST	NATMAINDEST	Main uses of dom com c nationally

Use the SUMMARY file to answer the following questions[9]:

What is the share of exports in GDP? [Header NGSM, Col share] *31%*

For which industry are Imported Intermediate inputs the biggest share of production costs? [CSTM, row share] *AirTrans 35%, Machines 31%*

In West Sumatra, which commodity do imports have the largest market share? [MSHR] *WaterTrans 53%*

Which three industries are most[10] labour intensive? [PCST, row share] *Paddy (97%), Crops (97%), GovServices (93%)*

Which region earns most from coal mining? [Header PCST, sum FAC, all DST] *East Kalimantan (94%)*

---

[8] Thus, even though TERMSUM.HAR is recreated each time you run a simulation, its contents will not change -- unless the input INFILE file was changed.

[9] You can sort the headers alphabetically to make it easier to find the headers. In ViewHAR, click on **File, Options, Sort headers alphabetically**.

[10] The little yellow (transpose) and green (sparse sorted) buttons at top left offer a quick way to answer "which is most" questions.

7.2 The Updated Data file

Open from ViewHAR the updated file *INFILE*. This file (actual name termsr.upd) contains postsimulation values for the same data items as are contained in the input *INFILE* file,

AGGMOD.HAR. Have a look at the header called 1CAP. Write down the new value for the East Kalimantan coal industry in the "updated" column below:

	Initial	Updated	Percent Change
CAP("Coal",EKalimantan”)	*19,477*	*19,358*	*-0.61*

Write down the original value of CAP("Coal",EKalinamtan”) in the "initial" column above (you were asked to note this previously). Then write down the percentage change between Initial and Updated, using the formula[11]:

%change = 100*(Final - Initial)/Initial

The CAP vector contains values of capital rentals. Each value is the product of a price, PCAP, and a quantity, XCAP. The solution file contains values (you were asked to note these previously) for percent changes in these two variables. Write in these two values below.

	pcap	xcap
CAP("coal",EKalinamtan”)	*-0.61*	*0*

Use the two variable results above to check that:

updated CAP = [original CAP]*[1+pcap/100]*[1+xcap/100]

updated CAP = *[19,477]*[1+(-0.61)/100]*[1+0/100] = 19,358.19*

How does GEMPACK know which price and quantity variables must be used to update CAP? That information comes from a line in the file TERM.TAB (near line 235), which reads:

```
Update (all,i,IND)(all,d,DST) CAP(i,d) = pcap(i,d)*xcap(i,d);
```

This updated data file has the same headers as the original input file. In fact, it is possible to use the updated data as the starting point (or initial data) for a second simulation. We might do this if we wanted to find the effect of one shock *followed by* another shock. Updated data files are used in a similar way by recursive dynamic models, a type of multiperiod CGE model which solves one year a time. The simulation for year T produces an updated data file which is used as input by the simulation for year T+1.

Close all the *.HAR, *.TAB, *.SL4 files before you begin with Part B.

[11] If you are too tired to do mental arithmetic, click the Programs menu item at the top of ViewHAR, then click on the Calculator icon which appears.

Part B: The operational phase of a new coal mine in Kalimantan

In Part B of this exercise we simulate the long-run impact of developing a new coal mine located in East Kalimantan. We model the long-run impact as an increase in the exports of East Kalimantan coal. New export markets are already secured, which will absorb the output of the new mine without affecting export prices received by existing mines. Assess the regional and national effects of the project, using a static version of IndoTERM

You will be working in the same working directory as before: ***C:\DTERMCRS\EX1***

8. View the long run closure

Several simulations can be carried out on the same model by changing the closure and/or the shocks as described in the Command file. For this exercise we are going to use the same model as described in TERM.TAB and used in Part A above.

Open the DOS box and type the following command:

```
tabmate termlr.cmf
```

Compare the short-run and long-run command files. An alternative way to compare to files in TABmate, is to open another TABmate program. You can then open two *.cmf files in two separate TABmate programs. In TABmate click on **File| Another TABmate.** In the new TABmate program, click on **Open** and search for termsr.cmf in the folder called **C:\ DTERMCRS\EX1**. You can now view these command files next to each other.

The first 59 lines are identical in the two command files. The differences occur in the closure swaps and shock statements. In the long-run closure, there are two sets of swap statements. The first 6 swap statements create the long-run closure and the last two swaps are policy-specific.

What do the following swap statements do? Does this confirm with your view of a long run closure?

```
swap         xcap = fgret;
swap flabsup_id   = xlab_id;
```

The first swap statement endogenises capital and the second swap statements exogenises national employment. Yes (capital is enogenous, employment exogenous)

The following shock exogenises the nominal trade balance to GDP.

```
swap   houslack   = shrBoTnom;
```

What does it mean when we fix the ratio of the trade balance to GDP?

It means that the balance of trade (X-M) as a share of GDP does not change. Household consumption will adjust to accomodate the restriction on the trade balance.

There are two policy specific swap statements. The first swap statement makes the export demand curve for East Kalimantan coal flat as to accommodate the new supply. We assume that the new coal exports to the world will not alter the price for coal. The second swap statement makes capital (xcap) in East Kalimantan coal exogenous. This allows us to shock this variable to account for the large investments undertaken in the construction phase.

There are two shock statements. The first shock statement increases capital stock by 20%. This increase represents the investment made in developing the coal mine, which now translates into an increase in capital stock.

Ecologia claims to have discovered new coal seams. To avoid simulating a scarcity of natural resources we need to increase the natural resource endowment by 20% to match the anticipated capital increase.

9. Run the simulation

After you have reviewed TERMLR.CMF, run the simulation. Type in the DOS box:

```
gemsim -cmf termlr.cmf
```

The message in the DOS box should show that the program has completed without error.

10. View the results

To view the results, type

```
viewsol termlr.sl4
```

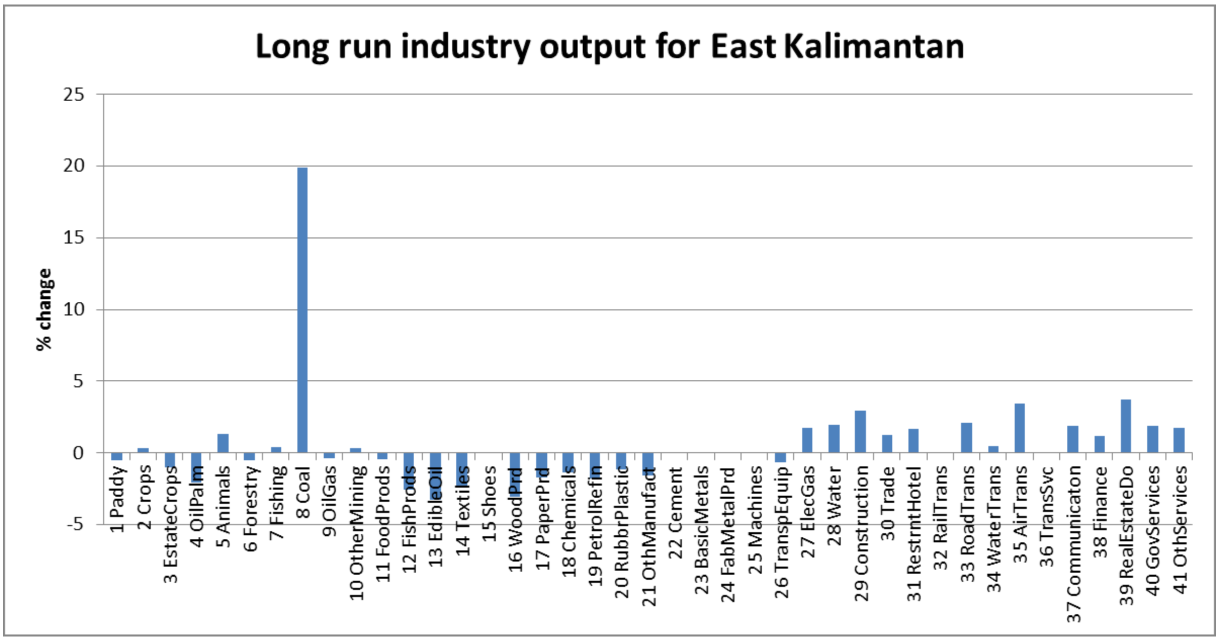
Examining values of the **NatMacro** variable. Complete column B (you have already completed column A in the previous section):

		Column A	Column B
	Main macro variable	**Short-run**	**Long-run**
1.	RealHou: aggregate household spending	*0.16*	*0.25*
2.	RealInv: aggregate investment	*1.22*	*0.32*
3.	RealGov: aggregate government spending	*0*	*0.22*
4.	ExpVol: aggregate exports	*-0.58*	*0.12*
5.	ImpVolUsed: aggregate imports	*0.42*	*0.23*
6.	RealGDP: real expenditure side GDP	*0.09*	*0.22*
7.	AggEmploy: aggregate employment	*0.16*	*0.00*
8.	AggCapStock: aggregate capital stock	*0.0*	*0.32*
9.	ExportPI: export price index	*0.11*	*0.15*
10.	CPI: consumer price index	*0.21*	*0.16*
11.	p1lab_io: average nominal wage	*0.21*	*0.34*
12.	realwage_io: average real wage	*0.00*	*0.17*

Find the MainMacro variable and examine regional-specific long-run outcomes. Complete the table below.

	Variable	East Kalimantan	North West Java
1.	RealHou: aggregate household spending	*2.71*	*0.00*
2.	Rea;Inv: aggregate investment	*2.75*	*-0.04*
3.	ExpVol: aggregate exports	*3.93*	*-0.69*
4.	ImpVolUsed: aggregate imports	*1.45*	*0.04*
5.	RealGDP: real expenditure side GDP	*2.89*	*-0.09*
6.	AggEmploy: aggregate employment	*1.27*	*-0.14*
7.	AggCapStock: aggregate capital stock	*2.71*	*-0.04*

Scroll down till you find the variable xtotsho (industry outputs). Double-click to view the numbers. Now right click on the column name "**6 EKalimantan**". A graph should appear that shows the percentage change in the output of industries located in East Kalimantan. Your graph may appear as a line graph. Change your graph to a bar graph by click on "**Vertical Bars**" in the left top corner. Your graph should look like the one below.



What do you notice? Identify the three best (apart from coal) and worst performing industries.

Best performing industries: RealEstateDo, AirTrans, Construction

Worst performing: EdibleOil, WoodPrd, FishProds. These industries are export orientated industries.

Services in general do well with Construction, Airtransport and RealEstateDo doing the best. We can assume that an increase in coal exports uses rail transport as a means of transportation.

Close all *.TAB, *.CMF, *.HAR and *.SL4 files before you continue with Exercise 2.